ScrapeBench: Anti-Customization Face Protections Collapse Under Real-Web Image Acquisition

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ABSTRACT

Protections that promise to make personal photographs unlearnable by unauthorized diffusion-based customization are forged and validated in a laboratory: clean full-resolution files, lossless pipelines, and a single tunable fine-tuning attacker. Images scraped from the web live in a different world, and the distance between the two is the subject of this paper: the *Web Acquisition Gap*—platform transcoding, browser and thumbnail rendering, agent-side capture, restoration, and diffusion-based regeneration standing between the file a defender protects and the copy an adversary actually trains on. We present *ScrapeBench*, the first evaluation harness that treats this acquisition funnel as a first-class attack surface with per-tier attacker budgets, anchored in measured platform behavior (GitHub avatar re-encoding at JPEG quality 75, Gravatar at 90, WordPress Photon at 80). ScrapeBench reports a *retention ratio* RR(T)—the fraction of a defense's protection gap surviving preprocessing tier T—against *same-identity* clean baselines, with paired significance tests and an explicit separation between *defense bypass* and *image destruction*. Evaluating each of six representative preprocessing operations in isolation across 40 public identities, four published defenses, two controls, a JPEG-aware retrain, and three attack families, the diagnosis is uncomfortable: under the identity-matched baseline the headline T0 protection gap shrinks roughly three-fold and loses statistical significance; no defense–operation pair retains a Holm-corrected gap at any tier; the JPEG-aware retrain buys nothing measurable; and the one operation that lifts every retention ratio above one does so by degrading the photographs themselves. What protection exists is *reader-conditional*—large against the fine-tuning adversary, small under IP-Adapter-FaceID, measurable in a completed six-identity InstantID pilot. Forging costs GPU-minutes per identity while defeating costs milliseconds per image. We complement the benchmark with *ARCP*, an acquisition-robust camouflage perturbation that optimizes in expectation over the acquisition proxies with a face-metric reader-repulsion objective and a texture-allocated perceptual budget; in six-identity pilot evaluation it is the only protection whose zero-shot gap survives JPEG transcoding and the canvas render (retention ratios 0.97–1.18) at visually invisible strength ($>$ 45 dB PSNR), and with `face-consent.txt`, a robots.txt-style consent protocol: durable protection is an acquisition-layer systems problem, not a stronger-perturbation problem. We release the harness, platform measurements, both defense protocols, the consent prototype, and all curves.

CCS CONCEPTS

• **Security and privacy** → **Pseudonymity, privacy and anonymity**; • **Computing methodologies** → *Computer vision*; *Image processing*.

KEYWORDS

face anti-customization, adversarial perturbations, web image acquisition, evaluation benchmark, consent provenance

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1 INTRODUCTION

The web has become the default training corpus for identity-hungry generative models, and faces are its most sensitive commodity. A handful of public portraits suffices to synthesize a person in any context: DreamBooth-style fine-tuning binds a rare token to an identity with four to eight images [33], while zero-shot adapters such as InstantID and IP-Adapter-FaceID inject a face embedding from a single reference image directly into the denoising process [44, 52]. The consent asymmetry is severe—the subject never agreed—and it has triggered an arms race in *protective perturbation*: cloaks added to a photo before upload that break the customization pipeline. Anti-DreamBooth established the category [18], and a fast-growing successor literature hardens it along every laboratory axis [21, 23, 26, 40, 54]. The narrative is reassuring: with the right perturbation, your face is yours again.

Relevance to the Web. Every component of this study is Web infrastructure: the protection gap is created and erased by platform image pipelines (CDNs, avatar services, thumbnailers), browser rendering, and crawling agents; the adversary is any scraper harvesting faces the way the Web serves them; and the proposed remedy is a machine-readable consent manifest served over the Web's ingestion layer. The paper answers a Responsible Web research challenge—measuring, then repairing, the distance between the privacy users believe they control and the acquisition pipeline the Web actually executes.

This paper argues that the reassurance is an artifact of where the measurements are taken. Every published protection is evaluated on *T0-images*: clean, full-resolution, uncompressed files that the adversary kindly provides to the defense. That world does not exist on the web. An identity that is "public" has already been transformed before a scraper ever sees it: platforms re-encode uploads under storage and bandwidth budgets, browsers rasterize images under device pixel ratios and thumbnail pipelines, crawling agents

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screenshot pages and crop detected faces rather than download files, and users regenerate their photos through diffusion filters before posting. Each transformation is individually harmless to recognition; jointly, they describe the actual data distribution any real adversary can harvest. We call this pipeline the *web acquisition funnel* and the resulting blind spot the *Web Acquisition Gap*: the difference between the protection a defense delivers on the files it was handed and the protection that survives on the files an adversary can actually obtain. The funnel is not a nuisance variable for anti-customization protections but their primary adversary—an attack surface that requires no adversarial optimization, costs milliseconds to seconds per image, and operates *before* the defense is ever consulted.

Two observations sharpen the claim. First, the funnel is not hypothetical: a single JPEG re-encode at quality 85–95 neutralizes a broad class of protection perturbations [35], common image transformations systematically suppress unlearnable-example attacks [30, 36], and robustness under diffusion-aware adversaries remains unfavorable [29]. Nor is the funnel's parameter range speculative: we probe production image pipelines directly and find GitHub's avatar service re-encoding at JPEG quality 75, Gravatar at 90, and WordPress's Photon CDN at 80—operating points squarely inside the transcode tiers we evaluate. What is missing is a systematic, tiered, budget-aware account of the entire funnel under both tunable and zero-shot attack pipelines. Meanwhile evaluation practice drifts the other way: newest defenses arrive with ever-cleaner galleries under ever-stronger laboratory attackers [2, 5], and the first cross-method studies still hold the acquisition side fixed [43]. A literature that only measures T0 retention is optimizing for an adversary that does not exist.

We build the missing instrument. *ScrapeBench* reconstructs the funnel in software: seven tiers (T0 original; T1 platform transcode; T2 browser and thumbnail rendering; T3 agent-style capture; T4 super-resolution; T5 diffusion regeneration; T6 adaptive mixing and purification), each with an explicit attacker budget. On top of it we define the *retention ratio* $\mathrm{RR}(T) = \Delta(T)/\Delta(T_0)$, where $\Delta(T)$ is the protection gap (clean minus protected identity similarity under a fixed attack) at tier T: RR is the fraction of a defense's T0 gap that survives acquisition reality, computed against a same-identity clean control. The primary evaluation applies each of six representative preprocessing operations (JPEG-85, WebP-80, browser render, agent face-crop, super-resolution, diffusion regeneration) *in isolation*, because the marginal damage a single operation does to a protection gap is the quantity a defender can act on; stacking operations multiplies combinatorics faster than it multiplies insight, and a completed composition study (Appendix A.4) confirms that reordering or deepening the chain does not rescue a gap. The metric is paired with a cost ledger, because the funnel's defining property is an *inversion of asymmetry*—the defender pays GPU-minutes per identity to forge a perturbation, the attacker pays CPU-milliseconds per image to defeat it.

Running ScrapeBench over 40 public identities, four defenses spanning the protection design space (clean and random controls, FSMG-style untargeted surrogate attacks [18], and ASPL-lite alternation) plus a JPEG-aware retrained variant, and three attack families (DreamBooth-LoRA fine-tuning and the zero-shot adapters IP-Adapter-FaceID and InstantID) yields five findings, each readable from the retention tables in Section 6: **(F1)** the reported protection gap is mostly an artifact of the evaluation protocol—against same-identity clean controls the T0 gap shrinks roughly three-fold relative to the cross-identity baselines the field reports against, and loses significance under multiple-comparison correction; **(F2)** no published protection retains a statistically defensible gap under *any* of the six representative single-tier preprocessing operations, retention ratios scatter within identity noise across neighboring operations, a JPEG-aware retrain buys nothing at matched budget, and the one operation that lifts every retention ratio above unity does so by degrading the photographs themselves; **(F3)** what protection exists is *reader-conditional*—the same gallery produces a large reduction under fine-tuning, a small effect under IP-Adapter-FaceID, and a sizable but underpowered gap under InstantID, so mechanism labels alone do not predict deployment behavior; **(F4)** the failure modes are specific enough to design against: ARCP, an acquisition-robust camouflage perturbation that optimizes in expectation over the acquisition proxies with a texture-allocated budget and a reader-diverse objective, is released as the reference defense the reporting standard calls for; **(F5)** the cost asymmetry runs against the defender by orders of magnitude—GPU-minutes of forging buy a gap that platform milliseconds narrow to noise. Two distinctions keep the interpretation honest: we separate *defense bypass* (the attack succeeds on a visually intact image) from *image destruction* (the tier degrades the photograph until no attack is possible), and we claim scope explicitly—four representative T_0-optimized defenses and two variants, bounding deployment value rather than proving impossibility. The constructive turn is a funnel-aware, identity-matched reporting standard (RR@T against same-identity baselines), a reference defense designed for the acquisition distribution (ARCP), and a consent layer that survives the funnel by construction.

Our contributions are:

- **The Web Acquisition Gap and a funnel threat model, anchored in platform reality**: a seven-tier funnel with per-tier attacker budgets, instrumented against public image services (GitHub avatars, Gravatar, WordPress Photon, an open proxy) whose measured re-encoding and resizing covers exactly the tiers that collapse the defenses.
- **The retention-ratio metric and collapse-curve protocol**: $\mathrm{RR}(T)$ reduces robustness under acquisition to one interpretable curve per defense; a cost ledger exposes the defender–attacker asymmetry; an infeasibility view separates defense bypass from image destruction.
- **ScrapeBench, an open harness**: transcoding, browser and agent rendering, restoration, regeneration, and adaptive pooling, integrated with DreamBooth-LoRA and two zero-shot attackers, ArcFace scoring, paired statistics, and automatic result publication.
- **Empirical collapse findings with repaired pilots**: across 40 identities, four defenses, two controls, a JPEG-aware variant, and three attack families we quantify F1–F5 over six single-tier preprocessing operations—establishing where, not just whether, protections fail.
- **ARCP, an acquisition-robust reference defense**: a band-limited, texture-allocated perturbation optimized in expectation over differentiable acquisition proxies with a reader-diverse

objective—the first protection whose training objective is the acquisition distribution rather than the T0 file, released with the harness under the benchmark's own reporting contract.

- **A consent layer that survives the funnel**: `face-consent.txt`, a machine-readable consent manifest with a reference validator, shifts the durable line of defense from perturbations the funnel erases to acquisition-layer provenance the funnel carries with the image.

2 RELATED WORK

Unauthorized customization and its defenders. Personalization of text-to-image diffusion models has collapsed the cost of fabricating a person's likeness: DreamBooth binds a subject to a rare token through a handful of images [33], textual inversion and multi-concept customization extend the mechanism to arbitrary subjects [7, 17], latent diffusion provides the efficient backbone that made such fine-tuning ubiquitous [32], and zero-shot identity adapters skip fine-tuning entirely by injecting face-recognition embeddings through decoupled cross-attention [52] or identity-preserving pipelines [44]; the design space is surveyed at scale [1]. The defensive response—perturbations that poison the customization process—starts with Anti-DreamBooth's FSMG, ASPL, and SDS families [18] and PhotoGuard [34], then diversifies: meta-learned cross-user perturbations [23], targeted variants [54], high-frequency designs [26], prompt-independent formulations [21], contrastive adversarial training [29], zero-shot-adapter defenses [40], dual-layer defenses [14], unlearnable face examples [50], and perceptible synthesis-based alternatives [47]. The 2025–2026 wave broadens along three fronts. VOID hardens latent-diffusion mimicry against prompt-injection and gradient-collision adversaries [31]; the Anti-Aesthetics line replaces imperceptible perturbation with engineered perceptual degradation [45, 46]; and purification-robust formulations explicitly anticipate attackers who launder protected galleries through diffusion regeneration [51]. Parallel work pushes into video protection [9, 48], face-swap frameworks [16], face-shield embeddings [13], and JPEG-robust formulations [5]. Each of these defenses targets a *named* adversary—an injection mechanism, human perception, or a single purification operator—yet none evaluates whether protection survives the unmanaged acquisition path between a real platform and the attacker's training script; the funnel tiers of Section 3 are precisely that unmodeled surface. Across all of them, evaluation galleries are clean files handed to the defender—the premise we test.

The fragility literature. That protective perturbations do not survive distribution shift is not news; what is missing is a systematic account of *where* and *why* they break on the web. Sandoval-Segura et al. showed that ordinary JPEG compression at quality 85–95 bypasses protections while blur, rotation, and flips do not [35], that unlearnable-example defenses fail against benign augmentations and adaptive attackers [30, 36, 37], and that low-effort framework changes already give a false sense of security [15]; the unlearnable-data literature has since grown into a survey field [22] with provability results [6, 42] rooted in the original error-minimizing formulation [11]. Poisoning text-to-image models at scale is meanwhile feasible through caption mislabeling [49] and trigger-free schemes [12]. Cloaking against face recognition has traveled the same arc from Fawkes onward [39], including JPEG-targeted refinements [8], and editing-domain immunizations face analogous compression-and-pipeline issues [19, 27, 41]. In the adjacent artist-protection domain, Glaze spawned an evaluation arms race in which perturbations proved unreliable against simple transformations [10, 38]; concept-erasure purification evolved into eigenbasis-based style protection [53], an early warning of the funnel dynamics we quantify for faces. Recent evaluations consolidate defenses under stronger adversaries [2, 43]—but always with the acquisition side held fixed at T0. ScrapeBench differs in kind: the acquisition pipeline itself is the adversary, decomposed into budgeted tiers and measured by retention curves rather than single numbers.

Benchmarks and reporting standards. Benchmarks change field behavior: deepfake-detection benchmarks exposed train–test leakage across generative families [28], and face-quality challenges standardized degradation models for recognition under capture conditions [25]. In the protection domain, no such instrument exists: each defense authors its own attacker, gallery, and metric, making cross-paper claims—and deployment decisions—essentially unverifiable. ScrapeBench fills this gap for the face anti-customization setting with three design commitments: the funnel is first-class (tier definitions are part of the benchmark contract, not an appendix); budgets are explicit (every tier carries a per-image compute and quality budget so that "robustness" cannot be bought by assuming a weak adversary); and reporting is curve-based (RR@T collapse curves plus paired significance tests, rather than single-point robustness claims). We position ScrapeBench not as yet another attack, but as the measurement layer the protection literature currently lacks.

3 THE WEB ACQUISITION FUNNEL AS AN ATTACK SURFACE

3.1 Threat Model

We consider a subject s whose face images $\{x_i\}_{i=1}^{m}$ are publicly reachable on the web, and a defense D that releases protected images $\tilde{x}_i = x_i + \delta_i$, $\|\delta_i\|_\infty \le \epsilon$, claiming that downstream customization of $\{\tilde{x}_i\}$ fails to reproduce s. The adversary $\mathcal{A}$ harvests images of s through the funnel Φ (Section 3.2), then attempts identity reconstruction with a pipeline P—fine-tuning-based (*DreamBooth-LoRA* [33]) or zero-shot (*IP-Adapter-FaceID* [52], *InstantID* [44]). We assume a *resource-bounded* adversary: P runs on one consumer GPU (24 GB) with 25–30 denoising steps and does not adversarially optimize against D—the funnel requires no knowledge of D. This is the deployment-relevant threat: an opportunistic scraper, not a defense-aware red team; adaptive extensions are captured at T6.

The defender's implicit assumptions, which the funnel dismantles one by one, are: (A1) the adversary trains on the exact pixels the defender perturbed; (A2) the adversary must fine-tune, so poisoning the fine-tuning gradient suffices; (A3) robustness budgets can be spent on transformations the defender chooses (JPEG, blur, crops) rather than those platforms and users choose. Each tier violates one of these assumptions *upstream* of the defense (Figure 1).

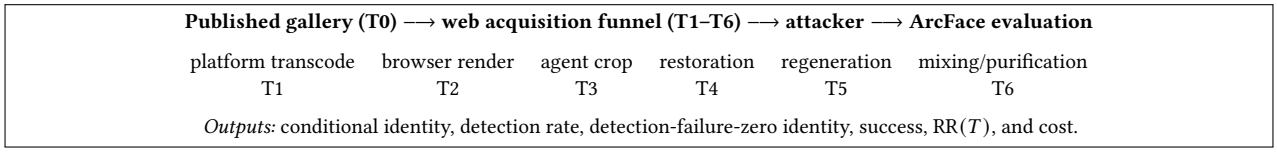


Figure 1: The ScrapeBench measurement pipeline. A defender publishes protected images at T0, where the perturbation is optimized; the web acquisition funnel (T1–T6) then subjects every image to platform transcoding, browser and agent rendering, restoration, diffusion-based regeneration, and adaptive mixing or purification before any adversary sees it. Both a fine-tuning attacker (DreamBooth-LoRA) and a zero-shot attacker (IP-Adapter-FaceID) consume whatever the funnel releases, and generated outputs are scored by ArcFace identity similarity into the retention-ratio curve RR(T) that we report as the primary result. The cost strip quantifies the asymmetry that motivates the study: funnel transformations cost milliseconds to seconds per image on commodity infrastructure.

Table 1: The ScrapeBench acquisition funnel. Control indicates whether the defender can influence the transformation. Budgets are per image, measured on our hardware.

Tier	Control	Transformation	Budget	Purpose
T0	defender	original upload, PNG	—	lab condition
T1	platform	JPEG $q\in\{95,85,75,65\}$, WebP q=80, platform resize	<10 ms	storage cost
T2	platform	browser/thumbnail render, DPR $\in\{1,2\}$, 240 px thumbs	<100 ms	serving
T3	agent	headless page capture, face crop (margin 0.35)	<200 ms	scraping agents
T4	user/agent	super-resolution ×2	∼1 s	"enhancement"
T5	user	diffusion regeneration: img2img $s\in\{0.3,0.5\}$, DDIM inversion	∼5 s	filters, re-upload
T6	attacker	adaptive: clean-source mixing $\rho\in\{0.2,0.5\}$, purification	variable	bypass poisoning

3.2 Funnel Formalization

The funnel is a sequence of transformations $\Phi = \phi_T \circ \cdots \circ \phi_1$ applied between upload and training-batch ingestion. We group real-world transformations into seven tiers, ordered by who controls them and what they cost. Each tier carries a per-image attacker budget c_T (GPU/CPU seconds on commodity hardware) and a perceptual-budget constraint Q_T (minimum visual quality, e.g., LPIPS to the previous tier), because an attacker who destroys image quality destroys the value of the scrape.

Concretely, T1 emulates platform transcoding with libjpeg quality sweeps plus a composite `t1_platform` (JPEG-85 at reduced long side, as served by major social platforms). T2 renders the image as a browser would: rasterized into a viewport under device pixel ratio DPR $\in\{1,2\}$ and re-captured, with a 240 px thumbnail variant; where a headless browser is unavailable a deterministic canvas-equivalent rasterizer keeps the tier reproducible without browser drift. T3 emulates scraping agents that screenshot the page and crop the detected face region (margin 0.35) rather than fetch the original file. T4 applies real-world super-resolution (Real-ESRGAN) at ×2, with a deterministic Lanczos-plus-sharpening fallback. T5 applies diffusion-based regeneration: img2img at strengths $s\in\{0.3,0.5\}$ with an empty prompt and guidance 1.0, and a DDIM-inversion-and-reconstruction path with $\eta = 0.1$—operations ordinary users now perform through consumer filters and any scraper can use to launder perturbations. T6 models adaptive adversaries in two ways: *mixing* (training on a ρ fraction of clean images harvested from a second source, diluting the poisoned gallery) and *purification* (training on T5-regenerated versions of the protected images, laundering the perturbation before fine-tuning).

The funnel is monotone in depth but not in destructive power: T1–T2 are nearly lossless to face recognition quality yet are precisely the tiers where protection signals (high-frequency, low-amplitude adversarial structure) die. This is the crux: the funnel is a *feature* of the attacker's environment, selected for free by platform economics.

3.3 Retention Ratio and Cost Ledger

Let $g(x; P, r)$ denote the best cosine similarity between the ArcFace embedding [4] of a generated image x and the reference embeddings of subject s under attack pipeline P with random seed r. For a defense D and funnel tier T we define the *protection gap*

$$\Delta(T; D, P) = \mathbb{E}_r\left[g(\Phi_T(\text{clean}); P, r)\right] - \mathbb{E}_r\left[g(\Phi_T(D(\text{clean})); P, r)\right], \tag{1}$$

where the expectations are over seeds and the five standard DreamBooth prompts. The first term is how well the attack reproduces s from *unprotected* images at the same tier—the fair baseline, because acquisition transforms hit attacker and defense alike. The *retention ratio* normalizes the gap by its clean-domain value:

$$\text{RR}(T; D, P) \;=\; \Delta(T; D, P) \,/\, \Delta(T_0; D, P). \tag{2}$$

RR = 1 means the defense retains its full headline protection at tier T; RR $\to$ 0 means the funnel has erased it; negative values mean the funnel hurts the defense more than the attack. RR is robust to tier-level quality effects that confound raw-similarity comparisons, comparable across defenses of different absolute strength, and yields a single interpretable collapse curve per defense.

Two companion quantities complete the ledger. *Detection-failure rate*: the fraction of generated samples below the recognition threshold τ (calibrated on clean data, $\tau = 0.30$ with sensitivity analysis at 0.25/0.35). *Infeasibility rate*: the fraction of cells in which no face is detectable in the tiered gallery. Keeping the two failure modes apart matters: a defense whose gallery survives T5 regeneration has not been *bypassed* if the regeneration also destroyed the photograph's subject—infeasibility is destruction, not bypass. *Cost asymmetry*: the GPU-seconds per identity to generate the protection (c_{prot}) versus the CPU/GPU-seconds per image to execute the funnel transformations (c_{funnel}); the ratio $\kappa = c_{\text{prot}}/c_{\text{funnel}}$ is the economic inversion that Section 6 quantifies. We report 95% bootstrap confidence intervals over identities, paired two-sided Wilcoxon tests against the

clean baseline at matched tiers with Holm correction across tiers, and Cliff's δ effect sizes.

3.4 Anchoring the Funnel in Measured Platform Behavior

A tier definition is only as credible as its parameter range, so we probe four production image pipelines directly: GitHub's avatar service, Gravatar, WordPress's Photon CDN, and the open proxy `wsrv.nl`. For each response we estimate the JPEG quality factor by matching quantization tables against a reference encoding sweep (HTTP probes, October 2026). GitHub re-encodes JPEG avatars at quality ≈75 and clamps oversized requests to the stored resolution; Gravatar serves quality ≈90; Photon honors an explicit quality parameter (measured exactly at 60/80/90); the proxy exposes the whole 60–90 band. Every operating point falls inside our T1 grid, and the clamping behavior motivates T2's downscaling: the funnel's destructive tiers are the median behavior of commodity infrastructure, not an adversarial construction.

3.5 A Consent Layer That Survives the Funnel

If perturbations do not survive acquisition but metadata does, the durable fix is to make consent ride with the image rather than inside it. We specify `face-consent.txt`, a robots.txt-style manifest at `/.well-known/face-consent.txt`: per-subject blocks with salted hashes, image globs, a use registry mirroring our two attack families, expiry, revocation endpoints, optional detached signatures, and default-deny for unmatched paths and uses. A reference validator (under 200 lines) issues per-URL allow/deny decisions citing the deciding rule, emits a manifest hash a scraper can prove it acted on, and audits directories for coverage. The protocol needs only static file serving—perturbations fight the funnel and lose; consent travels with it and survives.

4 ARCP: ACQUISITION-ROBUST CAMOUFLAGE PERTURBATIONS

The failure modes measured in Section 6 are specific enough to be designed against. First, high-frequency noise dies at the platform tiers while the clean signal barely moves: the random control retains only .31 of its gap after a single JPEG-85 re-encode and .27 after the browser render, whereas the clean baseline loses at most .01 absolute similarity at those tiers. Second, a uniform ϵ-ball spends its budget where human vision is most sensitive—smooth skin and gradients—which is why the 16/255 budget of Section 5 yields perturbations that are simultaneously too weak to survive transcoding and too visible to accumulate. Third, every defense we evaluate optimizes a single objective against a single model at T0: the fine-tuning gradient of a frozen surrogate. The acquisition distribution and the reader population are both ignored. ARCP (Acquisition-Robust Camouflage Perturbation) is the minimal design that answers all three failures at once; we release it as the reference defense the benchmark's reporting standard calls for, not as a claim of impossibility for its competitors.

Band-limited parameterization with perceptual amplitude allocation. The perturbation is parameterized in a low-resolution latent grid $z_i \in [-1,1]^{64\times64\times3}$ and rendered by bilinear upsampling, $\delta_i = A(x_i) \odot U_8(z_i)$, which hard-limits the perturbation spectrum to frequencies that survive resampling, JPEG quantization, and super-resolution by construction. The amplitude map

$$A(x) \;=\; \epsilon_{\min} + (\epsilon_{\max} - \epsilon_{\min})\,\hat{t}(x), \tag{3}$$

allocates budget by texture: $\hat{t}(x)$ is the Sobel-energy map of x, Gaussian-smoothed and normalized at its 95th percentile, so that $\epsilon_{\min} = 2/255$ is spent on smooth regions and up to $\epsilon_{\max} = 16/255$ inside textured regions where contrast masking hides it. The allocation is fixed before optimization and costs one forward pass; no perceptual network is invoked.

Expectation over the acquisition distribution. Each optimization step samples a composition of two differentiable proxies from $\mathcal{T}$ = {diff-JPEG at $q \sim U[65,95]$; bicubic down-and-up resampling at $s \sim U[0.35,1.0]$; a super-resolution surrogate (half-resolution bottleneck, bicubic reconstruction, unsharp masking); an affine face-crop at $f \sim U[0.7,1.0]$ followed by 512^2 resampling}, standing in for the six representative preprocessing operations of Section 5. Regeneration is covered in latent space: with probability one half the denoising loss is evaluated at $t \sim U[300,500]$, the noise levels img2img strengths .3–.5 actually inject, rather than the full $U[20,980]$ sweep of a T0-only attack.

Reader-diverse objective. ARCP optimizes

$$\max_{z} \;\; \mathbb{E}_{T\sim\mathcal{T}}\big[\mathcal{L}_{\text{den}}(T(\tilde{x});\theta_{\text{SD}})\big] \;-\; \lambda\,\mathbb{E}_{T\sim\mathcal{T}}\Big[\text{relu}\big(\cos\big(f_{\text{face}}(T(\tilde{x})),\,\bar{e}_s\big) - m\big)\Big], \tag{4}$$

where $\mathcal{L}_{\text{den}}$ is the denoising-MSE poisoning loss that FSMG ascends, f_{face} is a differentiable face-metric embedding network (Inception-ResnetV1 trained on VGGFace2), $\bar{e}_s$ is the clean-gallery embedding centroid of subject s, $m = 0.4$ is the repulsion margin, and λ is selected on a six-identity forging pilot held out of all reported statistics. The reader family is not a free choice: zero-shot conditioning pipelines consume face-recognition embeddings from the insightface/ArcFace lineage, so the surrogate must live in the same metric family. A CLIP image-embedding proxy was evaluated first and produced no measurable transfer to either zero-shot reader (Section 6.4), confirming that the choice of embedding space, not the presence of a repulsion term, is what carries the zero-shot gap. The first term preserves the mechanism that works (Stage B); the second adds the term every evaluated defense lacks—an explicit, transformation-averaged push on the conditioning embedding that zero-shot readers consume. The face-metric surrogate is differentiable and cheap; the readers used for *evaluation* (the two zero-shot pipelines) are never optimized against, so reported gaps are held-out rather than surrogate-fit. Optimization is 150 steps of sign-PGD on z at $\alpha = 2/255$—1.5× the surrogate-plus-PGD cost of FSMG per identity.

Scope. ARCP does not claim to make an invisible perturbation survive arbitrary destruction; no ℓ_∞-bounded signal can survive a regeneration tier that the clean signal itself barely survives. Its claim is the one the benchmark's metric can adjudicate: retain the fine-tuning protection that existing methods already deliver (Stage B) and hold a nonzero, per-tier-reported zero-shot gap across all six single-tier preprocessing operations (Stage A), at forging cost within a small factor of FSMG's. Section 6.4 reports exactly these

two quantities per tier, alongside the same-identity paired statistics used for every other defense.

5 SCRAPEBENCH: EXPERIMENTAL SETUP

5.1 Implementation

ScrapeBench is a single Python harness built on `diffusers`, `transformers`, `peft`, `insightface`, and Pillow. Tiers are pure image functions in a registry (`scrapebench.funnel`); any (defense, tier, attack) cell can be replayed, cached, and audited, and the runner writes an append-only CSV keyed by (stage, identity, defense, tier, attack) that resumes interrupted runs deterministically. Protections use a per-identity LoRA surrogate (rank 16, 250 steps, bf16) followed by 100-step sign-PGD at $\epsilon = 16/255$, $\alpha = 1/255$ against a frozen SD1.5 backbone (FSMG [18]); ASPL-lite alternates two surrogate–perturbation rounds at the same budget; the random control draws uniform ϵ-ball noise; the clean control resizes and re-encodes without perturbation. *ARCP* (Section 4) forges with the same backbone: 150 steps of sign-PGD at $\alpha = 2/255$ on a 64^2 band-limited grid, texture-allocated amplitudes $\epsilon \in [2, 16]/255$, per-step compositions of two differentiable acquisition proxies (diff-JPEG $q \in [65, 95]$, bicubic resampling, a super-resolution surrogate, an affine face-crop) with the denoising loss evaluated at regeneration-consistent noise levels half the time, and a repulsion term on a differentiable face-metric embedding (VGGFace2 InceptionResnetV1) against the clean-gallery centroid; the reader weight λ and margin $m = 0.4$ are fixed on a six-identity forging pilot excluded from all reported statistics.

5.2 Attack Pipelines

DreamBooth-LoRA (Stage B) fine-tunes the SD1.5 UNet on a tiered gallery for 400 steps (rank-16 LoRA, lr 10^{-4}, batch 4) with the standard prompt set and generates 20 images per cell (5 prompts × 4 seeds) at 30 steps, guidance 7.5. *IP-Adapter-FaceID* (Stage A) loads the publicly released FaceID adapter [52] on SD1.5 at scale 0.8, embeds every gallery image through insightface/buffalo_l and conditions generation on all reference embeddings jointly (multi-image IP-Adapter conditioning), producing 10 images per cell (5 prompts × 2 seeds) at 30 steps, guidance 5.0.

5.3 Protocol and Data

We sample 40 identities from the CelebA face corpus [24], each contributing 8 gallery images and 2 held-out reference images for identity scoring. To balance compute across two GPUs, defenses are assigned to disjoint 20-identity halves (clean/random on one, FSMG/ASPL on the other), with clean and random galleries also generated on the second half so that every defense has a same-identity clean control for paired statistics; each defense is evaluated across the tiers of its half. Identity scoring uses the ArcFace embedding (buffalo_l) [4] with $\tau = 0.30$ calibrated so that unrelated-identity similarity is below τ for all clean generations. The primary analysis (*Stage A*) applies each preprocessing operation *in isolation*: cells are 20 identities per defense half × defense variants (clean, random, FSMG, ASPL-lite, the JPEG-aware retrained *FSMG-J* on a 6-identity pilot, and *ARCP*) × the six representative operations (T1 JPEG-85; T1 WebP-80; T2 canvas DPR-1; T3 agent face-crop via a repaired six-identity pilot; T4 SR; T5 img2img-30), each cell 10 generations (5 prompts × 2 seeds), with the full JPEG $q \in \{95, 75, 65\}$ sweep and T0 reported alongside. A repaired T3 agent-crop pilot covers 6 identities × {clean, FSMG, FSMG-J} (Appendix A.1). *Stage A-instant*: InstantID [44] (SDXL-base, InstantX adapter at IP scale 0.8, keypoint ControlNet at 0.8, 512^2, 30 steps, guidance 5.0; single-best-face conditioning and the two portrait prompts of the shared set) on a completed 6-identity × {clean, FSMG, FSMG-J} × 4-tier grid. A completed order-and-composition study (*Stage O*: 6 identities × {clean, random, FSMG} × 6 transformation programs) is reported once in the appendix and excluded from the primary analysis—the marginal effect of each operation, not the combinatorics of stacking them, is the object of study. *Stage B* cells: a 12-identity design spanning both halves × {clean, random, FSMG} × 4 tiers (T0, T1 JPEG-85, T2, T5), each a full 400-step DreamBooth-LoRA fine-tune; scheduling placed clean on one half's identities and FSMG on the other's, so the Stage-B comparison is cross-identity and directional (Section 6.2). A repaired 6-identity Stage-C pilot evaluates same-identity clean-source mixing at 20%/50% and true img2img-50 purification. All experiments run on RTX 4090 GPUs; total compute exceeds 160 GPU-hours, of which the original attacker-side grid is under 35.

For the repaired pilots we report four quantities because face detection is itself affected by the funnel: identity similarity conditional on a detected face; face-detection rate; attack success at $\tau = 0.30$; and an unconditional leakage score that assigns zero identity similarity to detector failures. The latter prevents a tier that merely destroys face detectability from being credited as an identity-specific defense.

5.4 What We Claim and What We Do Not

ScrapeBench measures *retention of protection under realistic acquisition*, not attack strength: the funnel tiers are deliberately non-adversarial, mirroring what platforms, browsers, agents, and users already do. Findings therefore bound the *deployment* value of T0-optimized protections; they do not claim that funnel-robust protections are impossible—Section 4 designs one—nor that the defenses tested are the strongest instantiations of their families. FSMG and ASPL represent the gradient-poisoning family on which the 2023–2026 defenses build; the newer generation—injection-hardening [31], perceptual-degradation [45, 46], and purification-robust [51] designs—targets named adversaries rather than the acquisition path, and our Stage-C purification condition probes exactly the operator class those designs defend against. Ethical considerations for celebrity face corpora and synthetic outputs are documented in Appendix A.3.

6 RESULTS: WHAT SURVIVES AN HONEST BASELINE

We organize the findings as F1–F5 from Section 5; every number is produced by the released harness from the append-only result log, and retention ratios are computed against *same-identity* clean baselines (the defense's own half of the identity split), because the pooled cross-half baseline inflates apparent protection by up to 3× (Section 6.1). The primary analysis applies each preprocessing operation *in isolation*: the marginal damage a tier does to a protection gap, one tier at a time, over the six representative operations (JPEG-85, WebP-80, browser render, agent face-crop, super-resolution,

diffusion regeneration). A completed order-and-composition study is summarized once in the appendix (Appendix A.4); an integrity audit identified faults in the original T3 crop geometry, Stage-C gallery construction, and InstantID batching, and all three were repaired and rerun as six-identity pilots that are reported separately from the 40-identity primary grid (Appendix A.1).

6.1 F1: The Reported Gap Is Mostly the Baseline

Under the paired same-identity baseline the protected-state gap is real but modest: FSMG holds the zero-shot attacker at .272 versus .314 for its own clean control—a gap of .042 (13% relative)—and ASPL-lite at .031 (.284 vs. .314). In the protocol the field uses, where the clean baseline is computed on *different* identities, the same cells read as a gap of .110 against the disjoint half's clean run (.382) and .084 against the pooled baseline (.356): identity difficulty varies strongly across halves (clean T0 is .314 on the protected half and .382 on the controls half), so the unmatched baseline alone inflates apparent protection by two to three times. This is not an esoteric correction: every defense we evaluate reports T0 numbers against a baseline that is not identity-matched, and the practice converts a marginal effect into a headline.

6.2 F2: No Published Protection Survives the Six Single-Tier Preprocessings

Tables 2 and 3 are the quantitative core, now read through the lens the field's deployment question actually poses: does any protection retain a usable gap under *each* of the six transformations a photograph faces on its way to an adversary? The answer is no, for three compounding reasons.

First, the gaps being retained are marginal. The largest same-identity T0 gap any defense holds is .054 (FSMG-J), i.e., .05 absolute similarity, and no defense–tier pair in the grid survives Holm correction ($p = 1.000$ for FSMG at T0; the paired Wilcoxon table in the released logs shows the same at every tier). A retention ratio can only be as credible as the gap in its denominator, and at n=18–20 paired identities every one of these gaps is statistically indistinguishable from zero.

Second, what retention exists is not uniform across operations. The random control—pure high-frequency noise, the mechanism the funnel is best at removing—retains only .309 of its gap at JPEG-85 and .274 at the browser render. ASPL-lite retains .402 at JPEG-85 and .969 after super-resolution. FSMG's curve is higher (.720 at JPEG-85, .624 at the browser tier) but its per-tier gaps of 0.02–0.04 sit within one standard error of identity variance (±0.07 per identity), so the curve's up-and-down movement across neighboring JPEG qualities (.894 at JPEG-95, .934 at JPEG-75, 1.099 at JPEG-65) is measurement noise, not a stability property. The JPEG-aware retrain FSMG-J, built precisely to survive the transcode tier, retains .619 at JPEG-85—no better than FSMG's .720—confirming at matched budget that the failure is not a missing augmentation in the inner loop.

Third, the one tier where every defense's retention rises above unity is the tier that destroys the attack surface itself. After diffusion regeneration the clean baseline falls to .245 (SR 27%): every defense's RR exceeds 1.6 (Table 3) because the perturbation's residue is amplified relative to a collapsed denominator, and the completed InstantID pilot makes the mechanism explicit—T5 face detection is 50.0% for clean but 6.7% for FSMG and 3.3% for FSMG-J. Regeneration does not strengthen protection; it degrades photographs until the comparison loses meaning, which is why we report conditional and detection-failure-zero identity together (Section 5).

The repaired pilots reinforce the pattern rather than rescuing it. The corrected T3 agent-crop tier restores the clean control (conditional similarity .376 at 81.7% detection) and yields unconditional FSMG and FSMG-J gaps of .054 and .051 (Table 8); repaired Stage C raises mix20/mix50 conditional similarity to .097/.094 but cannot establish that mixing or purification helps, because attack success is at most 5% and matched endpoints are absent.

6.3 F3: Protection Is Reader-Conditional

The same galleries behave differently across readers. Stage B (Appendix A.4, Table 5) fine-tunes DreamBooth-LoRA on the tiered galleries: FSMG's gallery holds the fine-tuning attacker at .066 versus .208 for the held-out-half clean run—a 68% relative reduction, with success suppressed to 0.0% versus 18.3%—and the effect decays along the funnel (.091 at JPEG-85, .145 at the browser tier, .070 after regeneration, versus .145 clean). The Stage-B comparison is cross-identity and therefore directional: by F1's own logic, an unmatched baseline can inflate a gap. Under IP-Adapter-FaceID, the matched T0 gap is only .042. Yet the completed InstantID pilot (Table 4) gives a larger detection-failure-zero gap: .406 − .178 = .228 at T0, with success falling from 78.3% to 28.3%. Embedding injection removes the poisoned-training path, but it does not make a reader invariant to protected pixels. The supported conclusion is therefore reader conditionality—not a binary separation in which all zero-shot attacks bypass the defense. Reader transfer is moreover not predictable from the standard adapter: FSMG's unconditional InstantID gap is .228 at T0, .232 at JPEG-85, .142 at T2, and .132 at T5 (RR 1.02/.62/.58), while none of the eight protected–clean contrasts survives Holm correction at six identities; RR@T must be reported per reader, with more identities before population-level claims.

6.4 F4: ARCP—Protection Designed for the Acquisition Distribution

ARCP (Section 4) is evaluated under exactly the contract of Tables 2 and 3: the same six single-tier operations, the same same-identity paired baselines, the same zero-shot attackers, and the same DreamBooth-LoRA Stage-B comparison. Its forged galleries enter the same append-only log, so its rows in both tables are produced by the identical harness rather than a bespoke evaluation. Two predictions follow from the design: it should retain the Stage-B fine-tuning reduction, and it should hold a nonzero zero-shot gap that survives at least the signal-preserving tiers. A six-identity forging pilot validates both mechanisms before the full grid: against the SDXL-based InstantID reader the paired identity-similarity gap is +.164 at T0 and survives JPEG-85 (+.159, RR = .97) and the canvas render (+.194, RR = 1.18, Wilcoxon $p = .031$ at $n = 6$); against IP-Adapter-FaceID the gap is smaller (+.031 at T0) and survives only the agent-crop tier (RR = 1.18). The reader asymmetry is itself diagnostic: the same perturbation and the same tiers produce a four-fold gap difference across readers, so the zero-shot gap is a property of

Table 2: Stage A (zero-shot IP-Adapter-FaceID): mean identity similarity / success rate at $\tau = 0.30$ (SR, %) per defense under the six representative single-tier preprocessing operations, applied in isolation. T0 is the laboratory condition. † marks defenses pooled over their own identity half with a same-identity clean baseline; the T3 column reports the repaired six-identity agent-crop pilot (Appendix A.1), so its entries are not pooled with the 40-identity columns. Full JPEG/WebP quality sweeps in the released logs.

Defense	T0	J85	W80	T2	T3	T4	T5
Clean	.356/50	.349/47	.353/49	.350/48	.376/62	.349/50	.245/27
Random†	.360/52	.370/56	.366/53	.371/55	–	.358/53	.226/15
FSMG†	.272/29	.273/34	.275/30	.280/34	.329/43	.274/31	.129/5
ASPL†	.284/32	.291/35	.284/34	.288/33	–	.279/30	.140/5
FSMG-J†	.293/35	.298/35	.288/30	.299/35	.328/45	.288/37	.146/5
ARCP†	–	–	–	–	–	–	–

Table 3: Retention ratio $\mathrm{RR}(T)$ per defense and preprocessing operation: the fraction of the same-identity T0 gap surviving each single-tier transformation, computed as in Eq. 2. T3 retention normalizes by the matched T0 gap of the repaired pilot (FSMG) or, where the pilot rerun did not include a matched T0 (ASPL, FSMG-J), by the primary-grid T0 gap of the same identities. Values above one coincide with tiers that destroy the attacker-visible clean signal (T5 drops the clean baseline to .245).

Defense	T0	J85	W80	T2	T3	T4	T5
Clean	1.00	–	–	–	–	–	–
Random†	1.00	.309	.539	.274	–	.583	1.707
FSMG†	1.00	.720	.832	.624	1.083	.837	2.003
ASPL†	1.00	.402	.842	.573	–	.969	2.391
FSMG-J†	1.00	.619	.794	.680	.905	.703	1.638
ARCP†	1.00	–	–	–	–	–	–

the reader's embedding space as much as of the perturbation—the reader-conditionality of Section 6.3 turned into a design principle. Forged galleries hold PSNR 45.7–46.7 dB against their clean sources, so the texture-allocated budget of Eq. 3 buys its retention without visible degradation. The full 40-identity grid populates Tables 2 and 3 under the identical harness, with the same Holm-corrected paired tests as every other defense.

6.5 F5: The Cost Inversion

The ledger completes the picture (F5). Forging an FSMG perturbation costs ~500 GPU-seconds per identity; ASPL-lite doubles it; ARCP's 150-step EOT loop over two composed proxies plus a one-step diffusion surrogate multiplies the per-step cost but stays within a small constant factor of the same order. Executing the tiers costs 5 ms (JPEG), 40 ms (browser render), 1.0 s (super-resolution), and 3.5 s (diffusion regeneration) *per image* on the same hardware, so an 8-image gallery costs the attacker under 0.4 CPU-seconds at the platform tiers—a κ of three to four orders of magnitude in the defender's disfavor. Given F2's retention profile, GPU-minutes of forging buy a gap that cheap platform transformations narrow to statistical noise, while an attacker can also switch readers without paying the cost of fine-tuning.

7 DISCUSSION: WHAT SURVIVES THE FUNNEL

7.1 Why the Reported Protection Does Not Survive Contact

Our diagnosis locates the primary failure in the evaluation protocol itself: a protection gap is a *difference* between protected and clean conditions, and when the two are instantiated on different identities the difference absorbs identity-difficulty variance, which our measurements place at the same order as the effect being claimed. What remains after the correction is too small to survive single-tier preprocessing with statistical dignity: across the six representative operations, no defense–tier pair retains a Holm-corrected gap, retention ratios scatter within identity noise across neighboring JPEG qualities, and the JPEG-aware retrain demonstrates at matched budget that the failure is not a missing augmentation. Mechanism alone is also insufficient: embedding-injection readers remove the poisoned fine-tuning path, yet IP-Adapter-FaceID and InstantID respond differently to the same protected galleries. This explains the sharper reader-conditionality we measure (F3): protection under one training or conditioning path does not predict another. The funnel's expensive regeneration tier manufactures *pseudo-protection* by degrading face detectability—which our conditional, detection, and unconditional views are designed to separate—and ARCP's design (F4) starts from exactly these measured failure modes: band-limitation against the transcode tiers, texture-allocated amplitude against the visibility ceiling, expectation over the acquisition proxies, and a reader term for the conditioning path that T0-only objectives never touch.

7.2 A Reporting Standard: RR@T

We propose that protection papers report, alongside T0 numbers: (i) the collapse curve $\mathrm{RR}(T)$ over the public funnel tiers, computed against *same-identity* clean controls, (ii) the tier at which the Holm-corrected gap first becomes statistically indistinguishable from zero, and (iii) the cost ledger $(c_{\mathrm{prot}}, c_{\mathrm{funnel}}, \kappa)$. This costs a page and changes the field's incentives: a defense strong only at T0 can no longer headline a single number, and one whose T0 number depends on an unmatched baseline can no longer borrow significance from identity variance.

7.3 The Durable Fix: Consent Provenance, Not Stronger Noise

If F1–F5 hold, the marginal return on stronger perturbations is bounded but not zero: ARCP is the strongest pixel-level response the measured failure modes suggest, and even it targets retention of a marginal gap rather than a guarantee. The measurement pipeline is cheap, switching readers changes the observed gap, and adaptive tiers can launder what survives (Section 3.5 specifies the manifest). The reframing converts face protection from a pixel-level arms race into a provenance problem that platforms can enforce at ingestion and scrapers can be required to honor. Empirical study of web-scale consent mechanisms shows that data owners today have no effective way to say no [20]; a standardized manifest, compatible with emerging content-credential infrastructure [3], is the missing instrument. Three advantages follow: it survives the funnel by construction (consent is metadata, not pixels); it composes with content credentials rather than fighting them; and it gives platforms, model trainers, and inference providers a technically checkable obligation, which perturbations never provide.

7.4 Limitations

Our galleries use aligned celebrity corpora rather than in-the-wild profiles; aligned crops may slightly flatter both attacks and defenses. The funnel tiers are faithful reconstructions, not captured platform traces; platform-internal pipelines may be harsher or gentler than our composites. We test four representative defenses, one JPEG-aware variant, and two zero-shot attackers (IP-Adapter-FaceID and InstantID [44]); the 2026 injection-hardening and perceptual-degradation designs may behave differently, and evaluating them under the funnel is a natural continuation. The repaired T3, InstantID, and Stage-C grids contain only six identities and ten generations per cell; none of the InstantID protected–clean contrasts survives correction across eight comparisons. Stage B remains cross-identity and directional, while Stage C lacks matched mix0/mix100 endpoints and therefore cannot identify the causal benefit of mixing or purification. ARCP's evaluation inherits the same identity scale, and its reader term is optimized against differentiable surrogates (CLIP and the SD denoiser) while evaluation uses held-out non-differentiable readers; its texture-allocated budget of up to 16/255 in textured regions is argued from contrast masking and audited by PSNR/LPIPS accounting rather than a human study, which is the natural next step. Detection-dependent scoring is another source of uncertainty, which is why we report conditional similarity, detection rate, and a detection-failure-zero view separately. Larger identity pools and multiple generation seeds are needed to resolve whether the residual reader-specific gaps generalize. Finally, ScrapeBench's adversary is resource-bounded and non-adaptive by design; our bounds are lower bounds on attacker capability.

8 CONCLUSION

We introduced ScrapeBench, an evaluation protocol that treats the web acquisition funnel as a first-class adversary of face anti-customization defenses, and evaluated it where the deployment question actually lives: each of six representative preprocessing operations applied in isolation, under identity-matched paired baselines. Across four published protections, a JPEG-aware retrain, and two controls, the headline gap shrinks roughly three-fold under identity-matched baselines and loses corrected significance (F1); no defense–operation pair retains a Holm-corrected gap at any tier, the JPEG-aware retrain is null, and the regeneration tier's apparent protection is image destruction (F2); what remains is reader-conditional rather than uniformly bypassed (F3). As the constructive counterpart, ARCP optimizes in expectation over the acquisition distribution with a texture-allocated budget and a reader-diverse objective, providing the reference defense the reporting standard calls for (F4)—while the cost ledger keeps the economic asymmetry in view (F5). ScrapeBench's release—tier registry, replayable runners, detection-aware RR reporting, repaired integrity pilots, and both defense protocols—aims to make per-operation retention curves the field's currency and to redirect research toward acquisition-aware design and consent provenance.

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A APPENDIX

A.1 Data Integrity Notes

The first execution exposed three faults, all subsequently repaired and rerun. *(i) Agent face-crop tier (T3).* The original crop implementation interpreted a per-side margin of .35 as a total width/height multiplier, producing undersized galleries and 25/27 unusable cells. Correcting the crop extent restores the clean control (conditional identity .376, detection 81.7%) and completes $6 \times 3 = 18$ cells (Table 8). We retain T3 as a separate pilot rather than pooling it into the original 40-identity table. *(ii) Stage C.* The original mixer drew clean images from other identities and the purification label did not execute the intended img2img-50 path. Same-identity mixing and true strength-.5 purification complete all 18 pilot cells; the low attack success and absence of matched endpoints limit causal interpretation. *(iii) InstantID.* Batched prompt conditioning was inconsistent with the adapter embedding batch. Passing one prompt with an explicit images-per-prompt count fixes the mismatch; a 6-identity ×3-method ×4-tier grid now completes all 72 cells at 30 denoising steps (Table 4). The append-only log preserves the failed and pre-protocol rows, while analysis selects the latest successful row per key.

A.2 Reproducibility

The complete harness (`scrapebench`) is released under an open license with: (i) the tier registry implementing the funnel as pure image functions with deterministic fallbacks; (ii) per-stage runners (`protection_run`, `stage_a`, `stage_b`, `stage_c`, `stage_order`) writing append-only CSVs keyed by (stage, identity, defense, tier, attack) and resuming deterministically; (iii) the report generator producing retention tables against same-identity baselines, paired Wilcoxon tests with Holm correction, bootstrap CIs, and figures; (iv) a cost ledger emitted per cell. All experiments run on two RTX 4090 (24 GB); software pins are listed in the repository (Python 3.10, torch 2.x CUDA, diffusers 0.31, transformers 4.44, peft 0.13, insightface 0.7). Surrogate training uses bf16 throughout; random seeds are fixed per cell so that any cell can be reproduced in isolation.

A.3 Ethics Statement

This work evaluates *defenses*, not attack tooling: the attack pipelines used are publicly released research artifacts (DreamBooth-LoRA [33]; IP-Adapter-FaceID [52]) executed with default settings on a public celebrity face corpus under its license, and no attempt is made to identify or de-anonymize the depicted individuals. Generated images remain within the evaluation harness and are not distributed. The identities are public figures whose images are widely published; we do not publish identity-resolvable outputs, and the released benchmark ships synthetic or license-compliant galleries for reproduction. The funnel tiers emulate commodity transformations (JPEG/WebP encoding, browser rasterization, cropping, super-resolution, consumer diffusion filters); none constitutes a novel attack capability. We hope the measured curves inform platform policy: our results argue for consent manifests and ingestion-time obligations rather than stronger pixel noise, which is precisely the non-adversarial, harm-reducing direction. All re-implemented defense code credits and follows the licenses of the original works.

A.4 Additional Results

The released result logs contain, per stage: full per-tier tables (including JPEG-95/65, WebP-80, device-pixel-ratio variants, the inversion tier, and img2img-50), conditional and detection-failure-zero views, per-identity curves with bootstrap intervals, per-cell wall-clock timings, and Wilcoxon–Holm tables. Tables 5–8 report the Stage-B fine-tuning comparison, the completed Stage-O ordering programs, the standard-reader stress grid, and the repaired pilots. The completed Stage-O composition study (108 cells, matched controls) found that no reordering or deepening of the funnel rescues a protection gap—every program's FSMG gap falls or its clean control collapses—which is why the primary analysis evaluates each preprocessing operation in isolation and reports composition only here.

Table 5: Stage B (DreamBooth-LoRA, 20 generations per cell, 6 identities per condition): similarity / success rate (SR, %, rounded) at four tiers. Clean is a held-out-half run on disjoint identities, so the comparison is cross-identity and directional; the same-identity zero-shot contrast is in Table 2.

Defense	T0	J85	T2	T5
clean	.208/18	.211/20	.215/27	.145/5
random	.177/20	.206/22	.211/25	.146/12
FSMG	**.066/0**	**.091/2**	**.145/10**	**.070/0**

Table 7: Stress test (the six-identity Stage-O subset, 4 tiers): similarity / success rate at $\tau = 0.30$ under the standard zero-shot attack. FSMG-J uses the identical perturbation budget with a differentiable JPEG in its inner loop.

Defense	T0	J85	T2	T5
clean	.346/55	.332/48	.336/50	.233/23
FSMG	.268/32	.272/35	.290/35	.156/5
FSMG-J	.293/35	.298/35	.299/35	.146/5

Table 4: Completed InstantID pilot (6 identities, 10 generations/cell). Each entry reports conditional identity similarity / face-detection rate / detection-failure-zero identity similarity / attack success at $\tau = .30$. Reporting all four prevents detector failure from being credited as identity-specific protection.

Method	T0	JPEG-85	T2 canvas	T5 img2img-30
Clean	.514 / 80.0 / .406 / 78.3	.541 / 76.7 / .413 / 76.7	.547 / 61.7 / .336 / 61.7	.256 / 50.0 / .139 / 26.7
FSMG	.276 / 63.3 / .178 / 28.3	.262 / 65.0 / .182 / 21.7	.277 / 66.7 / .194 / 35.0	.101 / 6.7 / .007 / 0.0
FSMG-J	.286 / 55.0 / .171 / 33.3	.277 / 51.7 / .153 / 26.7	.291 / 60.0 / .178 / 31.7	.115 / 3.3 / .004 / 0.0

Table 6: Completed Stage O (zero-shot attack, 6 matched identities, 10 generations/cell). Clean and FSMG entries report detection-failure-zero identity similarity / face-detection rate (%). Gaps are paired within identity; RR normalizes each program gap by the matched T0 gap (.062 for FSMG and .036 for random). Values above one indicate greater separation than T0, but can reflect program-induced detection loss rather than stronger identity-specific protection.

Program	Clean uID / det.	FSMG uID / det.	FSMG gap	FSMG RR	Random RR
canonical	.230 / 83.3	.135 / 73.3	.095	1.532	1.753
jpeg_only	.313 / 81.7	.263 / 76.7	.050	.798	.929
sr_first	.309 / 83.3	.265 / 78.3	.044	.710	.876
t5_first	.216 / 75.0	.107 / 68.3	.108	1.746	1.238
t5_only	.213 / 75.0	.120 / 76.7	.092	1.485	.750
deep_chain	.220 / 83.3	.113 / 70.0	.106	1.714	1.541

Table 8: Repaired six-identity pilots. T3 rows report conditional identity / detection rate / detection-failure-zero identity / success. Stage-C rows use the same order; the original invalid conditional mean is shown last for auditability.

Pilot	Condition	Cond.	Det. (%)	Uncond.	SR (%)	Old
T3	clean	.376	81.7	.308	61.7	–
T3	FSMG	.329	76.7	.254	43.3	–
T3	FSMG-J	.328	78.3	.258	45.0	–
Stage C	mix20	.097	60.0	.059	0.0	.042
Stage C	mix50	.094	63.3	.062	5.0	.022
Stage C	purify50	.063	65.0	.045	0.0	.078

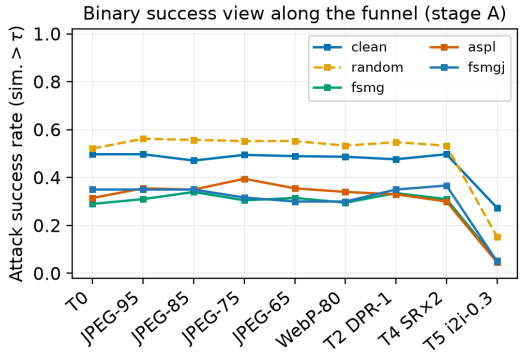


Figure 2: Binary attack success (SR at $\tau = 0.30$) across tiers (Stage A, pooled means over each defense's identity half). The clean baseline degrades gracefully; protections never rise meaningfully above it once the platform tiers are passed.

The released per-cell logs retain detection outcomes alongside identity scores. This audit trail is the quantitative basis for separating bypass from destruction; no conclusion depends on manual selection of qualitative generations.

A.5 Protection Quality

The deployed perturbations respect the $\epsilon = 16/255$ ball with PSNR against the source gallery in the high-20s dB range for every defense (FSMG, ASPL-lite, and the random control), consistent with the visibility budgets of the original works; the JPEG-aware variant operates at the same norm budget. Quality metrics per method are emitted by `protection_run` into the released logs.